QUARK–HAWKING BRIDGE

The Quark–Hawking Bridge

Volume II. Campaigns C1–C12: methods, results, the IVP-12 audit

Version 1.3.0

30 data tables · 116 figures · 154 tests

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Abstract

Volume II describes the twelve campaigns of QHB v1.3.0 — from the finite-volume gap (C1) to the asymmetry lock (C12) — and the twelve-pass independent audit IVP-12. For each campaign the method, the headline numbers, the claim statuses and the falsifiers are fixed; all data tables rebuild byte-identically via make all.

The architectural principle: a campaign is not an experiment but a deterministic measurement inside the pinned model class. Seeds are pinned in qhb/pins.py; every data table carries a SHA-256 in data/rebuild_hashes.json; nondeterminism is absent as a result (DETERM, honest-negative).

Special attention goes to the new v1.3.0 campaigns: C12 implements the asymmetry lock — an eta-phase map with the boundary exactly at eta_crit = 0.0908885, gap-filling dynamics with negative feedback, and the combinatorial bound on macroscopic accumulations. Together with the audit and the 30-claim Evidence Engine this forms the complete circuit of the no-free-quarks proof.

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Примечание: оглавление построено на кодах полей. После редактирования щёлкните по оглавлению правой кнопкой и выберите «Обновить поле», чтобы обновить номера страниц.

1. The reproducibility architecture

The package pipeline is a linear chain of deterministic steps: campaigns write JSON tables into data/; the figure script builds 116 PNGs (16 archetypes × RU/EN × light/dark plus 600-dpi hires); the IVP audit recomputes the headline numbers by independent methods; the twin control cross-checks the bilingual mirrors; pytest pins the numbers in 154 tests; and tools/rebuild_data re-runs everything and compares the SHA-256 of every table.

The unit of reproducibility is not approximately the same number but byte identity. Fixed seeds (N8) enter every stochastic procedure: the C5 Monte-Carlo, the C8 bootstrap, the C11 escape scan, the C12 fill dynamics. Floating-point operations run in a fixed order; sympy and mpmath deliver platform-independent results within the stated precision.

Table 1.1 — Package composition v1.3.0

Artifact	Count	Note
JSON data tables	31	data/*.json + SHA-256 registry
Standard figures	64	16 archetypes × 4 variants
Hires figures 600 dpi	52	13 archetypes × 4 variants
pytest tests	154	all pinned numbers
Audit passes	12	IVP-12, all PASS
Evidence Engine claims	30	5 statuses

2. Campaigns C1–C4: gap, string, temperature, detector

C1 (gap_finite_volume) scans a 9×9 grid over volumes 6..32 and couplings 2.80..4.56. All 81 cells OPEN; the minimum 0.174490 at (3.45625, 6). Audit pass IVP-01 recomputes the grid with inline arithmetic, bypassing qhb.core calls, and confirms agreement within 1e-12.

C2 (string_tension) measures the tension from the exact Wilson law: six loops, a log-slope, the maximum deviation from the law 5.6e-17. Audit pass IVP-02 checks the derivative identity: $-\partial \ln W/\partial(RT) = \sigma$ to 1e-9. C3 (bridge_thermal) scans T_c over the grid and verifies the spin coefficients to one hundred digits (IVP-03); the total correction at beta* is +11.28%.

C4 (detector_response) builds the detector response by convolution with a Gaussian resolution and fits the pure core on [0, 2.5]. In the tail zone (2.5, 3.5] all fifty residuals are positive — the runs test is degenerate. Audit pass IVP-04 checks the upper-bound consistency; the T5 result: the tail shape is unresolved, only the amplitude bound ≤ 0.00035 is claimed.

3. Campaigns C5–C8: three bodies, Y-strings, tunneling, bootstrap

C5 (free_three_body) combines analytics and Monte-Carlo: the capture threshold 0.991211 solves the gamma-tail equation exactly, and seed 777040 yields exactly 2433/10000 escapes. Audit pass IVP-05 keeps the z-score within three sigmas on every run; the extra deep-checks battery runs twenty independent seeds.

C6 (confined_spectrum) computes the Y-string spectrum through the Fermat point (Weiszfeld iteration, 200 rounds): the equilateral shape minimizes the energy. Audit pass IVP-06 proves the 120-degree condition symbolically — the dot products of the unit directions equal −1/2 exactly. C7 (tunneling_window) fixes the closed WKB window: S(0) = 13.8564, T(0) = 9.21e-13; the mpmath quadrature confirms the closed form with a 3.5e-16 fp64 deviation and below 1e-35 at high precision (IVP-07).

C8 (bootstrap) pins the statistics of the valley floor: 2000 resamples give a bias of 0.0046 and an error of 0.0124; the jackknife gives 0.0016, agreeing in sign and scale (IVP-08). The 95% interval [0.1695, 0.1955] covers the true floor — the robustness of the reference point is confirmed.

4. C9: the phase map and the naive-window falsification

C9 (phase_map) is the central gap map: 9 volumes × 41 couplings = 369 cells, all OPEN. The valley is refined parabolically from the three grid points around the argmin: beta* = 3.3583 and Delta_F,min = 0.181777 are recovered exactly because the profile is quadratic. Audit pass IVP-09 compares the parabolic refinement with golden-section search: the golden section stalls at the fp64 tie floor on a flat quadratic (~1e-8), so the honest pass level is 1e-6; the parabola delivers 8.4e-15.

The naive-window falsification is a standalone result, C9-FALS: direct measurement at the predicted closure point beta_c = 2.9910 gives 6.3 thresholds. The jackknife stress in deep_checks drops any three points around the valley and recovers beta* to better than 1e-9 — the refinement does not depend on the floor neighborhood.

5. C10: competing causal models

C10 (competing_models) decomposes the bridge signal into three causal channels: A — a symmetric Gaussian of width 3 (propagating), B — an antisymmetric braking channel (odd about t = 5), C — a pair of narrow contact echoes. An NNLS fit on a 1001-point grid recovers the shares 91.9509 / 8.0001 / 0.049 % with the additivity bias 1.3e-4, documented through the non-orthogonality scale 1.00014.

The pure-braking estimator B2 — the integral of the signal's odd part against f_B — vanishes on the even control with the residual 8.8e-18: a SymPy parity identity guarantees exact annihilation (C10-B2, exact; IVP-10 cross-checks NNLS against the normal equations). Out of window the model is invalid by construction: residuals grow as (t−t1)^4.071 — a fourth-order wing. This is the honest negative C10-NEG: no out-of-window claims are made.

6. C11: the vortex anchor

C11 (vortex_anchor) is the classical layer of the package. The Kirchhoff equations are implemented from scratch; an RK4 integrator with step 1e-4 reproduces the rigid rotation omega = 3/(2pi) with the deviation 2.3e-15 and the radius drift 3.3e-16. The linearization spectrum {0, 0, ±iomega, ±iomega} is derived by SymPy from the exact Jacobian; the characteristic polynomial $\lambda^2(4\pi^2\lambda^2+9)^2/(16\pi^4)$ matches identically (IVP-11).

The Aref run: a displaced triangle (a0 = 1.7, delta = 0.8627) with H = −0.374 < 0 lives for 30 time units; r_max = 2.7654 against the exact bound 2·sqrt(I) = 4.5075 and the virtual wall 12. The invariant drift over the run is 1.8e-15. The anchored barrier: $V(s) = V^*\cdot s^2\cdot e^{(2-2s)}$ with the maximum exactly at s = 1; 300 trajectories with energies 0.157..0.865 produce zero escapes.

7. C12: the asymmetry lock

C12 (asymmetry_lock) implements the principal new result. Nine perturbation profiles with unit sup-norm on [0, x*] give barrier losses from −2 (the saw control, strengthening the barrier) to exactly +2 (the step profile): the worst case is attained and proven (IVP-12, a SymPy inequality). The eta-phase map of 41 log-spaced points is open for all eta < 0.0908885 and closed beyond — the boundary passes exactly through eta_crit = Delta_F,min/2.

The gap-filling dynamics is a toy model of the observational route of T15. Two hundred wells, a million steps, the injection probability eta·1000: at eta_B a single hit per run and zero saturated wells (maximum fill 0.001); the positive control at eta = 0.5 injects on every step and saturates all wells in a cascade. The negative feedback is built in: filling raises the local closure cost, so the filling self-limits long before saturation.

The combinatorial bound completes the proof: with the per-attempt probability T(0) = 9.21e-13, the log-probability of a macroscopic accumulation of 1e30 quarks is −1.2e31. The experimental nugget-search bound (~1e-27 per nucleon) enters at the comparison status: it agrees with the zero prediction but does not prove the model. The route to falsification stays

open and written down: one CLOSED cell at eta ≤ eta_B, a resolved channel with T > 1e-12, or a nugget detection would refute the lock.

8. The IVP-12 audit and deep checks

The IVP-12 audit is twelve independent verifications: an inline-arithmetic recount of the C1 grid; the Wilson derivative identity; 100-digit spin constants; the tail upper-bound consistency; the C5 z-score; the symbolic Fermat 120 degrees; the WKB quadrature at 60 digits; the bootstrap-jackknife agreement; golden section versus the parabola; normal equations versus NNLS; the symbolic vortex polynomial and the zero equilibrium residual; the symbolic worst-case lock inequality and the 50-digit margin arithmetic. Verdict: 12/12 PASS.

Deep checks extend the audit: 100-digit recomputation of omega, $pi^2/8$, $pi^5/704$, eta_crit and the margin; the C9 jackknife stress; twenty C5 seeds with a z-score guard; the symbolic location of the barrier maximum; the symbolic parity identities of C10; byte-level determinism (31 tables, zero drift); the twin control of 14 pairs × 3 sweeps = 51 checks. Together these form the circuit in which every number of the package has an independent second derivation.

Table 8.1 — Headline numbers v1.3.0

Quantity	Value	Pinned in
beta* / Delta_F,min	3.3583 / 0.181777	N2, C9-VALLEY, tests
Detector threshold	0.0363485	N3, C1, tests
P_esc (free three-body)	0.2433 ± 0.0043	C5, seed 777040
omega (vortices)	3/(2pi) = 0.477464829	N1, T11, IVP-11
V* (barrier)	0.9359375	N7, C11, 0/300
eta_crit / margin	0.0908885 / 1.49e8	T14, IVP-12, 50 digits
Shares A/B/C	91.95 / 8.00 / 0.049 %	C10-SHARES
Out-of-window power	4.071	C10-NEG

9. Evidence Engine: 30 claims

The claims registry is the administrative core of honesty. Thirty claims in five statuses (exact ×7, measured-law ×12, honest-negative ×9, hypothesis ×1, comparison ×1) cover all theorems, all campaigns, the audit and reproducibility. Each claim carries a bilingual statement, a falsifier, a five-field uncertainty budget and pointers to the data tables.

The registry rules are stricter than intuition suggests: a hypothesis is never cited as a result; external values never prove the model; a negative is a full-fledged result; a claim

without a falsifier is not admitted. Such a registry turns the package from a collection of computations into a verifiable system of knowledge: any reader sees not only what is claimed, but what would refute it and where it is recorded.